

Physics-inspired coarse-scale perturbation

Phase 3 ablation: target the parameters at wavelengths $\lambda \gtrsim 3000$ km (upper synoptic and above). A different mechanism per model, same physical objective.

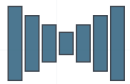
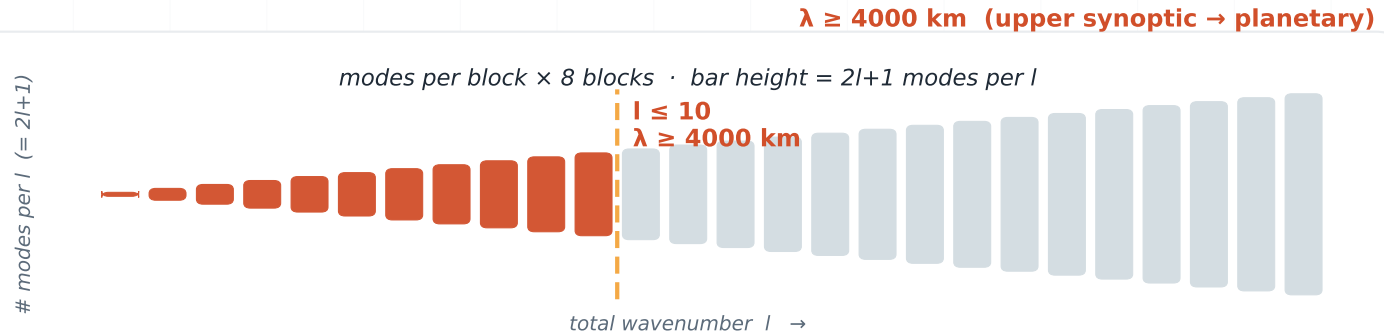


SFNO

Spherical Fourier Neural Operator

Mechanism:
spectral weight sub-slice perturbation

Target slice:
 $l \leq 10$ modes of every block's filter weight

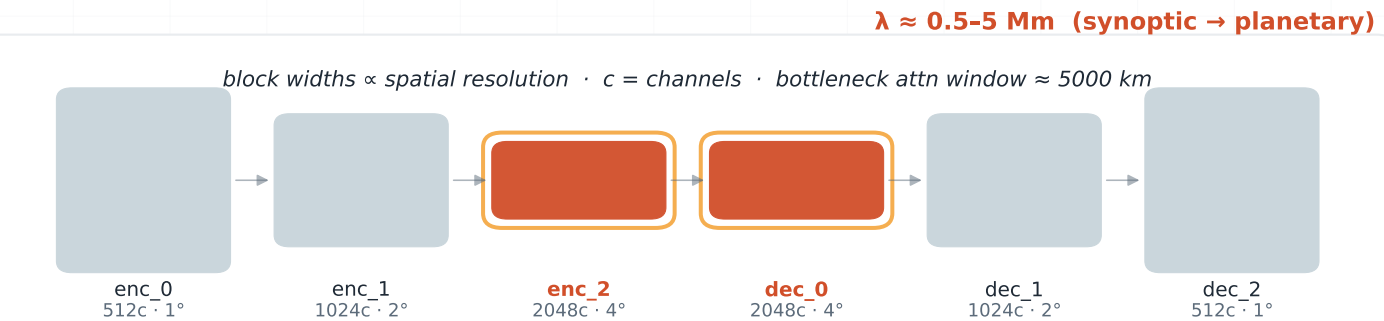


Aurora

Swin-Transformer 3D U-Net

Mechanism:
bottleneck-layer weight perturbation

Target slice:
encoder_layers.2 + decoder_layers.0

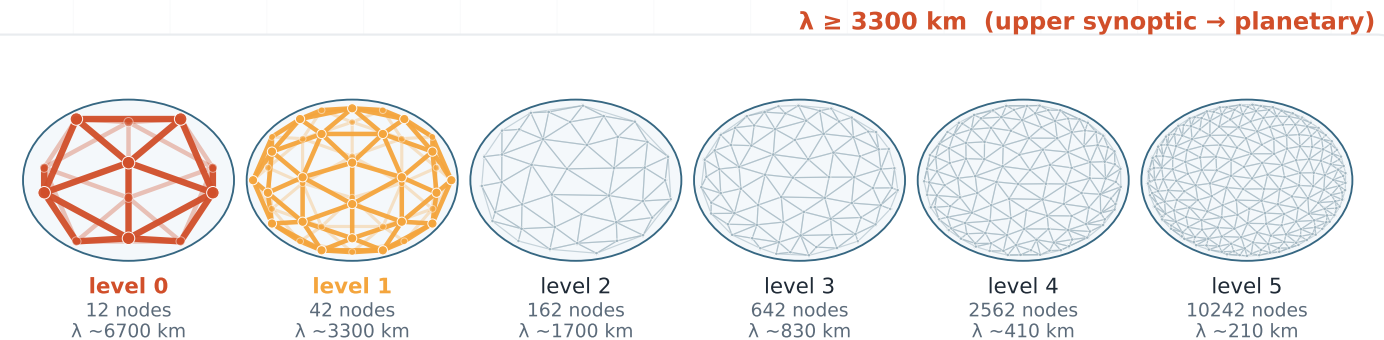


GraphCast

Multi-Mesh Graph Neural Network

Mechanism:
runtime hook on edge embeddings

Target slice:
edges at mesh refinement levels 0-1



● targeted (perturbed) ● intermediate scale ● untouched (small-scale)